caseSetup-v4.1.1 Technical User Guide

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Jan 2026

This document covers the use of caseSetup for external automotive aerodynamics cases.

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Contents

1 Introduction

`caseSetup` was built over a couple years for the Ontario Tech Racing team. This process is a mix of experiences gained from FSAE related activities and professional activities. The first iteration of `caseSetup` was born in the fall of 2022. The drive behind `caseSetup` was to improve the consistency of runs from user to user while reducing the chances of mistakes. The main idea behind it was to use pre-defined templates which mildly restricts the user from making un-intended modifications to the setup. This allows for consistent meshing, solving, and post processing regardless of user.

This document here is the technical user guide, while some Computational Fluid Dynamics (CFD) theories may be included, it is not meant to be an exhaustive explination or discussion of the physics. Please refer to specialized CFD books and resources if more information is required.

2 Directory Structure

This process requires a specific directory structure for your cases to work. It looks like the following:

- `JOB-DIRECTORY`
  - 01_incoming(YYYY-MM-DD)
    - 2025-09-11_New_Geometry
  - 02_references
    - ANSA
      - trial009.ansa.gz
    - MSH
      - trial009.obj.gz
      - fw-v1-1.obj.gz
  - 03_reports
    - 051_064_063_report_2025-11-16.pptx
  - CASES
    - 001
    - 002
    - 003
    - ...
    - 009
    - **caseSetup**
      - constant
      - system

The definitions for each folder is described here:

- `01\_incoming(YYYY-MM-DD)` - Contains all the incoming geometry from other departments or other files which are required for the simulation. Each instance of CAD delivered to the department must be in a dated folder with a short descriptive name.
- `02\_references` - Contains all the geometry files that are used in the simulation. Subdirectories are ANSA and MSH
- `ANSA` - Contains all the ANSA files that are used in the simulation, ideally contains the CAD files within them and not just the tri-surfaces. Should be saved as ".gz" for smaller file size.

- `MSH` - Contains all the tri-surfaces that are used in the simulation. Ideally this is where the original copies of all the STLs or OBJ files live. Each trial folder contains only a symbolic link referencing to this directory to reduce the number and size of geometry files.
- `03_reports` - Contains all the reports placed by the `pptGeneration` process or placed by the user manually.
- `CASES` - Contains all the simulation runs, this is where each case/trial folder lives, each trial is denoted by a 3 digit number XXX.
- `caseSetup` - This is the file that controls the simulations and the outputs.
- `system` - This directory contains the files required for the simulation to run.
- `constant` - Like `system` this directory also contains the files required for the simulation and also in `constant/triSurfaces` is where the geometry for the simulation is stored for this case. It is ideally a symbolic link so that repeated geometries are not copied many times in many cases.

Once all these directories are setup, we are ready to explore the rest of the features.

3 Basic Setup

To start, you would need a case directory created in your CASES directory. Once created, you will not have a caseSetup file just yet. To get a file, execute the following line in your case directory,

```
caseSetup --new
```

A fresh caseSetup file will be generated. The inside of the file will look like this:

```
[DEFAULTS]
DEFAULT_MODULES = defaultGlobalControl defaultGlobalSimControl defaultBCSetup defaultGlobalDecomposition
     defaultGlobalMaterial defaultGlobalRefinement defaultPost
ADDON_MODULES =

[TITLES]
CASENAME = casename
CASEDESCRP = default description
JOBCODE = TS

[GEOMETRY]
GEOM = default.stl,1,8,5,1.1,high
```

The DEFAULT_MODULES line controls the settings that are available. If the setting is not visible, it just means that it will use the default setup. Each setting here is explained below:

- CASENAME - the case name for your case, it does not do anything at this time, but in the future it could be added to the post processing images.
- CASEDESCRP - case description, see CASENAME
- JOBCODE - Two letter initial for the job you are running, just helps identify the job that's running on the cluster queue.
- GEOM - List of geometries to use in the case. A comma separates each setting for that particular geometry, the next geometry goes on the subsequent line.
  - What each column represents:

    ```
    GEOMETRY_FILE_NAME,SCALE,MESH_LEVEL,INFLATION_LAYERS,INFLATION_LAYER_GROWTH_RATE,
         WALL_MODEL
    ```

Each geometry that is required in this run and is in its own file should be listed here, the settings shown above will be applied to all geometries in that file. If a different settings are required to be applied, that geometry should be broken out into it's own file. A common example is wheels.

3.1 Rotating Geometries

The wheels and tires are spinning and therefore should have a rotating wall boundary condition applied, therefore it MUST be in its own part. Example,

```
[DEFAULTS]
DEFAULT_MODULES = defaultGlobalControl defaultGlobalSimControl defaultBCSetup defaultGlobalDecomposition
     defaultGlobalMaterial defaultGlobalRefinement defaultPost
ADDON_MODULES =

[TITLES]
CASENAME = casename
CASEDESCRP = default description
JOBCODE = TS

[GEOMETRY]
GEOM = body.obj.gz,1,8,5,1.1,high
        ROTA-fr-wh-lhs,1,8,5,1.1,high
        ROTA-fr-wh-rhs,1,8,5,1.1,high
        ROTA-rr-wh-lhs,1,8,5,1.1,high
        ROTA-rr-wh-rhs,1,8,5,1.1,high
```

Notice the prefix ROTA, it lets caseSetup know that this geometry will have a rotating wall boundary condition applied. Upon running caseSetup, if it's the first time with this set of geometry, caseSetup will stop and let you know that the following blocks have been added. The user should check that these are the appropriate settings for each wheel geometry.

NOTE: If using a plinth, ensure that the plinth is a separate geometry, else the automatic wheel center calculation will include the plinth in the radius calculation resulting in an incorrect radius and angular velocity!

```
[ROTA-fr-wh-lhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-fr-wh-rhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-rr-wh-lhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-rr-wh-rhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True
```

- `WH_RAD` [[`float,m`]] - wheel radius, distance from wheel center to ground (`r`, `default`)
- `WH_CENTER` [[`float,m`]] [[`float,m`]] [[`float,m`]] - wheel center coordinates in global coordinates system (`x y z`, `default`)
- `WH_AXIS` [[`float,m`]] [[`float,m`]] [[`float,m`]] - wheel axis of rotation from, using right hand rule with respect to the global coordinate system (`x y z`, `default`)
- `WH_AXIS` [`bool`] - activate rotating wall or use no-slip condition (`true`, `false`)

Using `default` option will trigger the automatic calculation but manual input of values is still possible. To speed up `caseSetup` it is recommended to copy the values output by the calculation in the termial and input them into the caseSetup if coordinates and axis values are not expected to change for subsequent runs.

3.2 Controlling Run Settings

Deleting the module `defaultGlobalControl` will expose the following block:

```
[GLOBAL_CONTROL]
STARTFROM = latestTime
STARTTIME = 0
ENDTIME = 1000
DT = 1
WRITEINT = 500
PURGEINT = 1
AVGSTART = 500
```

- `STARTFROM` [`str`] - which time to start the simulation at, `latestTime` will start the simulation at the lastest available time if running for an existing solution, `startTime` will have the simulation start at the value specified in `STARTTIME` (`latestTime`, `startTime`)
- `STARTTIME` [`float/int,s`] - time to start the simulation at, will be used if `STARTFROM` is specified as `startTime`, must use `int` if steady run is used, or `float` if a transient run is selected
- `ENDTIME` [`float/int,s`] - final time or time step to end the simulation at, must use `int` if steady run is used, or `float` if a transient run is selected
- `DT` [`float/int,s`] - timestep size, must use `1` if steady run is used, or any `float` if a transient run is selected

- `WRITEINT` [`float/int,s`] - interval which the data is written to disk for restart or post-processing use, must use `int` if steady run is used, or any `float` if a transient run is selected (**CAUTION:** saving too frequently can cause increased solve time and increased storage usage!)
- `PURGEINT` [`int`] - number of written timesteps to keep, will only keep `PURGEINT` number of time steps (aside from `STARTTIME` or `0`)
- `DT` [`float/int,s`] - when to start averaging flow variables, must use `int` if steady run is used, or `float` if a transient run is selected

3.3 Controlling Simulation Settings

Deleting the module `defaultGlobalSimControl` will expose the following block:

```
[GLOBAL_SIM_CONTROL]
SIM_TYPE = steady
SIM_INIT = potential
SIM_SYM = half
TURB_MODEL = kosst
INIT_P = default
INIT_K = default
INIT_OMEGA = default
INIT_NUT = default
INIT_NUTILDA = default
```

- `SIM_TYPE` [`str`] - what type of simulation (`steady`, `transient`)
- `SIM_INIT` [`str`] - what type of initialization, `steady` only available when `SIM_TYPE` = `transient` (`none`, `potential`, `steady`)
- `SIM_SYM` [`str`] - simulation symmetry type, reccomended to run in `half` if not doing yaw and vehicle is mostly symmetrical, use `full` if running internal flow (`half`, `full`)
- `TURB_MODEL` [`str`] - turbulence model applied, `transient` runs can only use `sa` ([`kosst`,`sa`])
  - `steady`
    - `kosst` - $k-\omega$ Shear Stress Transport ($k-\omega$ SST)
    - `sa` - Spalart-Allmaras
  - `transient`
    - `sa` - Spalart-Allmaras Delayed Detached Eddy Simulation (SADDES)
- `INIT_P` [`float,Pa`] - initial conditions for pressure, reccomended to leave in default unless specific values are required (`float`, `default`)
- `INIT_K` [`float,m2s-2`] - initial conditions for turublent kinetic energy, reccomended to leave in default unless specific values are required (`float`, `default`)
- `INIT_OMEGA` [`float,s-1`] - initial conditions for ω term, reccomended to leave in default unless specific values are required (`float`, `default`)
- `INIT_NUT` [`float,m2s-1`] - initial conditions for ν_t term, reccomended to leave in default unless specific values are required (`float`, `default`)
- `INIT_NUTILDA` [`float,m2s-1`] - initial conditions for $\tilde{\nu}$ term, reccomended to leave in default unless specific values are required (`float`, `default`)

3.4 Boundary Conditions

Deleting the module `defaultBCSetup` will expose the following block:

```
[BC_SETUP]
INLET_MAG = 15
YAW = 0
DRAG_VEC = default
LIFT_VEC = default
PITCH_VEC = default
GROUND = moving
REFCOR = 0.775 0 0
REFLEN = 1.55
REFWIDTH = 1
REFAREA = 1
POROSITY = yes
BASE_CELL_SIZE = 1.6
DOMAIN_SIZE = 48 16 16
DOMAIN_OFFSET = 0.5
FRT_WH_CTR = 0 0 0
```

- `INLET_MAG` [`float,ms-1`] - inlet velocity magnitude ([`float`])
- `YAW` [`float,deg`] - yaw angle, will only yaw in the positive direction, if `SIM_SYM` is `half`, this value not used (`float`)
- `DRAG_VEC` [[`float,m`] [`float,m`] [`float,m`]] - drag force vector, following SAE J1594 convention by default (`x y z`, `default`)
- `LIFT_VEC` [[`float,m`] [`float,m`] [`float,m`]] - lift force vector, following SAE J1594 convention by default (`x y z`, `default`)
- `PITCH_VEC` [[`float,m`] [`float,m`] [`float,m`]] - pitch moment vector, following SAE J1594 convention by default (`x y z`, `default`)
- `GROUND` [`str`] - ground plane boundary condition (`moving`,`stationary`)
- `REFCOR` [[`float,m`] [`float,m`] [`float,m`]] - point defining center of rotation, typically center of wheel projected to the ground (`x y z`)
- `REFLEN` [[`float,m`]] - reference length, typically wheelbase length (`float`)
- `REFWIDTH` [[`float,m`]] - reference width, typically trackwidth length (`float`)
- `REFAREA` [[`float,m`]] - reference area, typically frontal area or `1` to have calculations done in $C_D A$ (`float`)
- `POROSITY` [`str`] - to calculate forces including forces due to porous media, recommend to leave at `yes` (`yes`, `no`)
- `BASE_CELL_SIZE` [[`float,m`]] - base cell size of blockMesh, resulting mesh levels are calculated based off this number, recommend to leave at `1.6` (`float`)
- `DOMAIN_SIZE` [[`float,m`] [`float,m`] [`float,m`]] - dimensions for blockMesh domain (`x y z`)
- `DOMAIN_OFFSET` [[`float,m`]] - distance of `FRT_WH_CTR` to the inlet boundary as a ratio of domain `x` length, i.e. `0.5` means the `FRT_WH_CTR` is located at the center of the domain (`float`)
- `FRT_WH_CTR` [[`float,m`] [`float,m`] [`float,m`]] - coordinate for the center point projected to the ground of the front wheel axle (`x y z`)

3.5 Decomposition and Cluster Control

Deleting the module `defaultGlobalDecomposition` will expose the following block:

```
[GLOBAL_COMPUTE_SETUP]
QUEUE_TYPE = slurm
SCRIPT_VERSION = 1
NUM_NODES = 1
TASK_PER_NODE = 72
DECOMP = 18 2 2
```

- `QUEUE_TYPE` [`str`] - NOT ACTIVE, FOR FUTURE USE (`str`)
- `SCRIPT_VERSION` [`int`] - NOT ACTIVE, FOR FUTURE USE (`int`)
- `NUM_NODES` [`int`] - NOT ACTIVE, FOR FUTURE USE (`int`)
- `TASK_PER_NODE` [`int`] - number of cores to use (`int`)
- `DECOMP` [`int int int`] - decomposition pattern, dictates the number of divisions in each direction during decomposition, freestream direction should be the largest, the product of the 3 values must match `TASK_PER_NODE` (`x y z`)

3.6 Fluid Properties

Deleting the module `defaultMaterial` will expose the following block:

```
[GLOBAL_MATERIAL]
NPHASE = 1
MATERIAL_TYPES = air
VISCOSITY = 1.46e-5
```

- `NPHASE` [`int`] - NOT ACTIVE, FOR FUTURE USE (`int`)
- `MATERIAL_TYPES` [`str`] - NOT ACTIVE, FOR FUTURE USE (`str`)
- `VISCOSITY` [`float, m2s-1`] - fluid kinematic viscosity (`float`)

3.7 Meshing Control

Deleting the module `defaultGlobalRefinement` will expose the following block:

```
[GLOBAL_REFINEMENT]
TEMPLATE_TYPE = snappyTemplate1
MIN_GEOM_LEVEL = 8
REF_FEAT_ANGLE = 25
GEOM_REF_DIST = 0.01 0.05 0.1
GEOM_REF_LEVELS = 9 8 7
DEFAULT_WAKE_REF = defaultWake
LOC_IN_MESH = -1 -1 4
LAYER_TYPE = absolute
DEF_EX_RATIO = 1.2
REF_ANGLE = 25
FEAT_EDGE_ANGLE = 140
FEAT_EDGE_LEVEL_INC = 1
FIRST_LAYER_HEIGHT = 0.0001
FINAL_LAYER_HEIGHT = 0.001
FINAL_LAYER_RATIO = 0.3
```

- `TEMPLATE_TYPE` [`str`] - meshing template to use (`str`)
  - `snappyHexMesh` - template file must exist in `setupTemplates/<caseSetup_template>/defaultDicts/`
  - `ansaMesh` - `TEMPLATE_TYPE` string must include `caseSetup` template name as well as `ansa` separated by underscores i.e. `otr_ansa_default`. The `ansa` is used to trigger the use of `ansaMesh`. The template must be located in `setupTemplates/<caseSetup_template>/defaultANSA/`.
    The last value entry in `TEMPLATE_TYPE` is the name of the template within that path and can be of any length. Just NO SPACES!

- MIN_GEOM_LEVEL [int] - coarses mesh level to be used for all geometries in the case (int)
- REF_FEAT_ANGLE [float,deg] - angle where curvatures with angles larger than it would trigger mesh refinement in that area, operates only for ansaMesh (float)
- GEOM_REF_DIST [[float,m] [float,m] [float,m] ...] - distance from the surface which the mesh level from GEOM_REF_LEVELS shall be applied, unlimited entries, but must match length in GEOM_REF_LEVELS (float float ...)
- GEOM_REF_LEVELS [int int ...] - refinement levels for each corresponding distance in GEOM_REF_DIST unlimited entries, but must match length in GEOM_REF_DIST (int int ...)
- DEFAULT_WAKE_REF [str] - wake refinement generated by default or use specific template (true, false, str)
  - if using specific template, it should exist in setupTemplates/<caseSetup_template>/defaultRefinements/
- LOC_IN_MESH [[float,m] [float,m] [float,m]] - point which defines the "inside" location of the domain, only works for snappyHexMesh (float float float)
- LAYER_TYPE [str] - how the layer growth occurs globally, only works for ansaMesh, defined in snappyHexMesh by template (absolute, aspect)
  - absolute - first layer height is defined absolutely based on FIRST_LAYER_HEIGHT, rest of inflation layer sizing determined by aspect ratio set for the geometry
  - aspect - first layer height is defined as a ratio to cell size outside of the inflation layers, global settings currently hardcoded as 0.5, rest of inflation layer sizing determined by aspect ratio set for the geometry
- DEF_EX_RATIO [float] - global layer expansion ratio (float)
- REF_ANGLE [float,deg] - angle where curvatures with angles larger than it would trigger mesh refinement in that area, operates only for snappyHexMesh (float)
- FEAT_EDGE_ANGLE [float,deg] - angle under which triggers edge refinement, only works for snappyHexMesh (float)
- FEAT_EDGE_LEVEL_INC [int] - refinement level increment for edges defined by FEAT_EDGE_ANGLE, only works for snappyHexMesh (int)
- FIRST_LAYER_HEIGHT [[float,m]] - first layer height, only works for ansaMesh (float)
- FINAL_LAYER_HEIGHT [[float,m]] - total height of inflation layers, not currently operational, for future use (float)
- FINAL_LAYER_RATIO [[float,m]] - ratio of last layer and the surrounding element size, not currently operational, for future use (float)

3.8 Post Processing Control

Deleting the module defaultPost will expose the following block:

```
[GLOBAL_POST_PRO]
ISOQ = 2000
ISOCPT = 0
SLICES = True
XSLICE = default
YSLICE = default
ZSLICE = default
```

- ISOQ [float,s-2] - Q criterion value of which to generate VTK iso surfaces of (float)

- ISOCPT [float] - $C_{p,total}$ value of which to generate VTK iso surfaces of (float)
- SLICES [str] - to generate VTK slices (true, false)
- XSLICE [[float,m] [float,m] [float,m]] - start, end and spacing of the x normal slices (float, float, float)
- YSLICE [[float,m] [float,m] [float,m]] - start, end and spacing of the y normal slices (float, float, float)
- ZSLICE [[float,m] [float,m] [float,m]] - start, end and spacing of the z normal slices (float, float, float)

3.9 Ride Height Control

Deleting the module defaultRideHeight will expose the following block that is under development currently. The goal is to create a process which can do ride height mapping.

```
[RIDE_HEIGHT_SETUP]
RUN_RIDE_HEIGHT = false
RIDE_HEIGHT_FILE =
USE_DEPEND = True
```

4 Prefix Controls

There are 3 or 4 letter prefixs that allow special control of your geometry and simulation. These help with assigning porous media regions, Multiple Reference Frame (MRF) regions, or even more control over meshing for certain geometries.

4.1 Rotating Geometries

The `ROTA` prefix allows the assignment of rotating wall to geometries such as wheels. A usage example is shown below.

```
[DEFAULTS]
DEFAULT_MODULES = defaultGlobalControl defaultGlobalSimControl defaultBCSetup defaultGlobalDecomposition
     defaultGlobalMaterial defaultGlobalRefinement defaultPost
ADDON_MODULES =

[TITLES]
CASENAME = casename
CASEDESCRP = default description
JOBCODE = TS

[GEOMETRY]
GEOM = body.obj.gz,1,8,5,1.1,high
        ROTA-fr-wh-lhs,1,8,5,1.1,high
        ROTA-fr-wh-rhs,1,8,5,1.1,high
        ROTA-rr-wh-lhs,1,8,5,1.1,high
        ROTA-rr-wh-rhs,1,8,5,1.1,high
```

Notice the prefix `ROTA`, it lets `caseSetup` know that this geometry will have a rotating wall boundary condition applied. Upon running `caseSetup`, if it's the first time with this set of geometry, `caseSetup` will stop and let you know that the following blocks have been added. The user should check that these are the appropriate settings for each wheel geometry.

NOTE: If using a plinth, ensure that the plinth is a separate geometry, else the automatic wheel center calculation will include the plinth in the radius calculation resulting in an incorrect radius and angular velocity!

```
[ROTA-fr-wh-lhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-fr-wh-rhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-rr-wh-lhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True

[ROTA-rr-wh-rhs]
WH_RAD = default
WH_CENTER = default
WH_AXIS = default
ROT_WH = True
```

- `WH_RAD` [[`float,m`]] - wheel radius, distance from wheel center to ground (`r`, `default`)
- `WH_CENTER` [[`float,m`]] [[`float,m`]] [[`float,m`]] - wheel center coordinates in global coordinates system (`x y z`, `default`)
- `WH_AXIS` [[`float,m`]] [[`float,m`]] [[`float,m`]] - wheel axis of rotation from, using right hand rule with respect to the global coordinate system (`x y z`, `default`)
- `WH_AXIS` [`bool`] - activate rotating wall or use no-slip condition (`true`, `false`)

Using `default` option will trigger the automatic calculation but manual input of values is still possible. To speed up `caseSetup` it is recommended to copy the values output by the calculation in the termial and input them into the caseSetup if coordinates and axis values are not expected to change for subsequent runs.

4.2 Porous Media

When radiators are used in a simulation, it would be very computationally expensive to model each radiator fin. So instead we use the Darcy-Forcheimer porous media calculation to model the pressure drop across a radiator which allows us to estimate the mass flow across the radiator and resulting forces.

The prefix `POR` is used to define the porous media geometry. **CAUTION:** The porous media geometry must only contain the "box" or region that represents the radiator fin region, do not include the frame or any other part of the radiator. The box which represents the fin region should be a simple prism. The inlet and outlets should have their own PID with the keyword `-inlet` or `-outlet`. There should only be 1 of each PID!

The mass flow calculations are performed on each pid in the porous media, therefore including the walls or any other surfaces where flow isn't moving *across* would make the calculations erronous.

Including the prefix `POR` on a geometry would expose the following block:

```
GEOM = trial005-rad.obj.gz,1,8,6,1.4,high
       fw-H05A00.obj.gz,1,8,6,1.4,high
       rw-R11I01.obj.gz,1,8,6,1.4,high
       POR-rad-001.obj.gz,1,10,6,1.4,high
       ROTA-fr-wh-lhs-v3-3.obj.gz,1,8,6,1.4,high
       ROTA-fr-wh-rhs-v3-3.obj.gz,1,8,6,1.4,high
       ROTA-rr-wh-lhs-v3-3.obj.gz,1,8,6,1.4,high
       ROTA-rr-wh-rhs-v3-3.obj.gz,1,8,6,1.4,high

[POR-rad-001]
DCOEFFS = 1000
FCOEFFS = 1000
VEC1 = 0.7071 0 -0.7071
VEC2 = 0 1 0
POINT = 1.7875 -0.000583 0.342739
SAMPLE_POR = True
```

the name of the block will be the name of the geometry just without the extension to identify it.

- `DCOEFFS` [`float`] - Darcy coefficient attained from experimental testing of pressure drop vs flow rate of the radiator (`float`)
- `FCOEFFS` [`float`] - Forcheimer coefficient attained from experimental testing of pressure drop vs flow rate of the radiator (`float`)
- `VEC1` [[`float,m`] [`float,m`] [`float,m`]] - vector defining the normal of the radiator exit (`float float float`)
- `VEC2` [[`float,m`] [`float,m`] [`float,m`]] - vector in plane with the radiator exit (`float float float`)
- `POINT` [[`float,m`] [`float,m`] [`float,m`]] - point defining the "inside" of the radiator (`float float float`)
- `SAMPLE_POR` [`str,m`] - to sample the mass flow through the radiator (`true`, `false`)

4.3 GEOMX Controls

Using the `GEOMX` prefix allows the user to have more control over the meshing of a specific geometry. This is useful for geometries that need to have different meshing criteria than the `GLOBAL_REFINEMENT` settings.

Including the prefix `GEOMX` on a geometry would expose the following block:

```
GEOM = GEOMX-trial005-rad.obj.gz,1,8,6,1.4,high
       fw-H05A00.obj.gz,1,8,6,1.4,high
       rw-R11I01.obj.gz,1,8,6,1.4,high
       POR-rad-001.obj.gz,1,10,6,1.4,high
       ROTA-fr-wh-lhs-v3-3.obj.gz,1,8,6,1.4,high
       ROTA-fr-wh-rhs-v3-3.obj.gz,1,8,6,1.4,high
       ROTA-rr-wh-lhs-v3-3.obj.gz,1,8,6,1.4,high
       ROTA-rr-wh-rhs-v3-3.obj.gz,1,8,6,1.4,high
```

```
[GEOMX-trial005-rad]
GEOMX_MIN_MAX_LEVEL = 9 10
GEOMX_REFINEMENT_ANGLE = 25
GEOMX_FEAT_EDGE_LEVEL_INC = 1
GEOMX_FEAT_EDGE_ANGLE = 140
GEOMX_NLAYERS = 5
GEOMX_LAYER_RATIO = 1.2
FIRST_LAYER_HEIGHT = 0.0001
GEOMX_REF_DIST = 0.03 0.1 0.2
GEOMX_REF_LEVELS = 9 8 7
```

- `GEOMX_MIN_MAX_LEVEL` [`int int`] - minimum and maximum mesh level to be used for a specific geometry (`int int`)
- `GEOMX_REFINEMENT_ANGLE` [`float,deg`] - angle where curvatures with angles larger than it would trigger mesh refinement in that area, operates only for `ansaMesh` (`float`)
- `GEOMX_FEAT_EDGE_LEVEL_INC` [`int`] - number of levels to increase for mesh refinement for edges captured by `surfaceFeatureExtract` (`int`)
- `GEOMX_REF_DIST` [[`float,m`] [`float,m`] [`float,m`] ...] - distance from the surface which the mesh level from `GEOMX_REF_LEVELS` shall be applied, unlimited entries, but must match length in `GEOMX_REF_LEVELS` (`float float` ...)
- `GEOMX_REF_LEVELS` [`int int` ...] - refinement levels for each corresponding distance in `GEOMX_REF_DIST` unlimited entries, but must match length in `GEOMX_REF_DIST` (`int int` ...)
- `DEF_EX_RATIO` [`float`] - global layer expansion ratio (`float`)
- `REF_ANGLE` [`float,deg`] - angle where curvatures with angles larger than it would trigger mesh refinement in that area, operates only for `snappyHexMesh` (`float`)
- `FEAT_EDGE_ANGLE` [`float,deg`] - angle under which triggers edge refinement, only works for `snappyHexMesh` (`float`)
- `FEAT_EDGE_LEVEL_INC` [`int`] - refinement level increment for edges defined by `FEAT_EDGE_ANGLE`, only works for `snappyHexMesh` (`int`)
- `FIRST_LAYER_HEIGHT` [[`float,m`]] - first layer height, only works for `ansaMesh` (`float`)
- `FINAL_LAYER_HEIGHT` [[`float,m`]] - total height of inflation layers, not currently operational, for future use (`float`)
- `FINAL_LAYER_RATIO` [[`float,m`]] - ratio of last layer and the surrounding element size, not currently operational, for future use (`float`)